

CLINICAL PHARMACOKINETICS

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AN OVERVIEW

- The term *Pharmacokinetics* is derived from the ancient Greek words “*pharmakon*” means “drug” and “*kinetikos*” means “putting in motion”.
- Pharmacokinetics is the study of the time course of a drug’s movement in the body as affected by the absorption, distribution, metabolism and elimination. It is the understanding of what body does to the drug.

AN OVERVIEW

- Clinical Pharmacokinetics is the application of pharmacokinetic principles to the safe and effective therapeutic management of drugs in an individual patient.
- It helps to enhance the efficacy and decrease the toxicity of a patient's therapy.

NEED OF PHARMACOKINETICS

- Individualize patient drug therapy.
- Monitor medication with narrow therapeutic index.
- Decrease the risk of adverse effects while maximizing pharmacologic response of medications.
- Evaluate PK/PD as a diagnostic tool for underlying disease.

PHARMACOKINETIC PRINCIPLES

- Absorption
- Distribution
- Metabolism
- Elimination

ABSORPTION

- Absorption is the movement of the drug from its site of administration in to the circulation.
- Not only the fraction of the administered dose that gets absorbed, but also the rate of absorption

ABSORPTION

Factors affecting absorption are:

➤ Aqueous solubility:

Drugs given in solid form must dissolve in the aqueous biophase before they are absorbed.

For poorly water soluble drugs rate of dissolution governs rate of absorption.

A drug given as watery solution is absorbed faster than when it is given as solid form or as oily solution.

ABSORPTION

- Concentration:
- Area of absorbing surface:
- Vascularity of the absorbing surface:
- Route of administration:

ABSORPTION

- Drugs can be administered by a variety of routes. The choice of appropriate route in a given situation depends both on drug as well patient related factors.

Routes can be broadly divided into those for ;

- Local action
- Systemic action

ABSORPTION

➤ Local routes:

Topical, Deeper tissues, Arterial supply

➤ Systemic routes:

➤ Oral

➤ Sublingual or buccal

➤ Rectal

➤ Cutaneous

➤ Inhalation

➤ Nasal

➤ Parenteral – Subcutaneous

Intramuscular

Intravenous

Intradermal

ABSORPTION

Absorption in oral route

Nonionized lipid soluble drugs are readily absorbed from stomach as well as from the intestine.

Acidic drugs are predominantly unionized in the acid gastric juice and are absorbed from the stomach. Basic drugs are largely ionized and are absorbed only on reaching the duodenum.

Presence of food dilutes the drug and retard the absorption.

DISTRIBUTION

- Membrane Permeability
cross membrane to site of action.
- Plasma protein binding
bound drugs do not cross membranes
malnutrition – albumin = free drug
- Lipophilic drugs accumulate in adipose tissue.
- Volume of distribution.

METABOLISM

- Drugs can undergo metabolism in the lungs, blood and liver.
- Body works to convert drugs to less active forms and increase water solubility to enhance elimination.
- Liver – primary route of drug metabolism.

METABOLISM

- Liver may be used to convert pro- drugs (inactive) to an active state.

- Types of reactions.

Phase I (Cytochrome P450 system)

Phase II

ELIMINATION

- Pulmonary – expired in the air
- Bile – excreted in feces
enterohepatic circulation
- Renal – glomerular filtration
tubular reabsorption
tubular secretion

BASIC PARAMETERS

- In pharmacokinetics body is represented as a single or multiple compartments into which the drug is distributed.

Volume of distribution

Clearance

Elimination rate constant

Half life

Bioavailability

AUC

BASIC PARAMETERS

Volume of distribution

It is defined as the distribution of a medication between plasma and the rest of the body after oral or parenteral dosing.

Gives information on how the drug is distributed in the body.

Used to calculate the loading dose.

BASIC PARAMETERS

Clearance

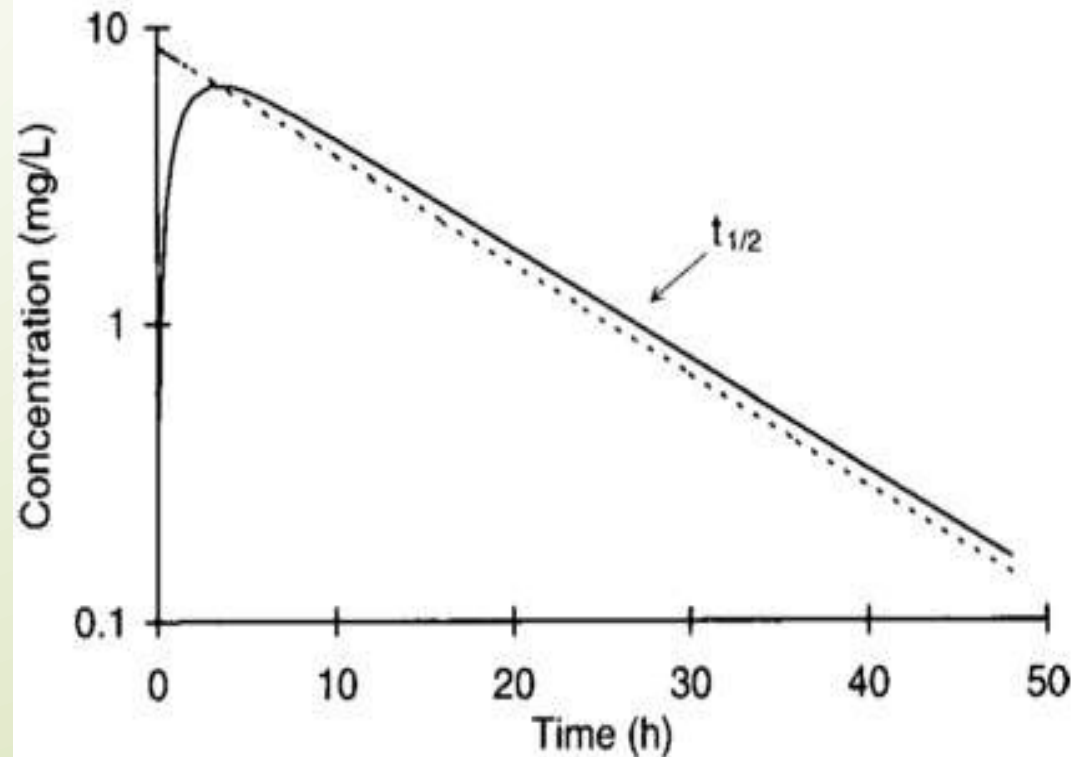
- Ability of organs of elimination like kidney, liver to clear drug from the bloodstream.
- Volume of fluid which is completely cleared of drug per unit time. Units are in L/hr or L/hr/kg
- Used in determination of maintenance doses.
- Maintenance dose will be in mg/hr so for total daily dose will need multiplying by 24.

BASIC PARAMETERS

HALF LIFE – $t_{1/2}$

- The time taken by the serum concentrations to decrease by its

➤ $t_{1/2} = 0.693$
 K_e

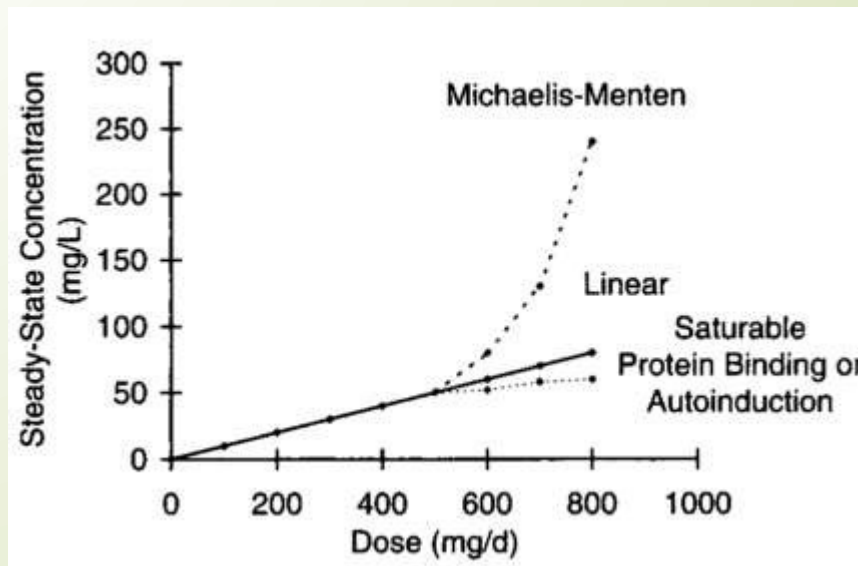


BASIC PARAMETERS

STEADY STATE

- Its defined as the amount of drug administered is equal to the amount of drug eliminated within one dosing interval resulting in a constant serum level
- Drugs with short half life will attain steady state rapidly, whereas drugs with long half life will take days to week to attain steady state.

LINEAR & NON LINEAR PHARMACOKINETICS



LINEAR & NON LINEAR PHARMACOKINETICS

- ▶ When doses are increased for most drugs, steady-state concentrations increase in a proportional fashion leading to *linear pharmacokinetics*.
- ▶ When steady-state concentrations change in a disproportionate fashion after the dose is altered, drug is said to follow *nonlinear pharmacokinetics*.
- ▶ Steady-state concentrations increase more than expected after a dosage increase, the most likely explanation is that the processes removing the drug from the body have become saturated.

LINEAR & NON LINEAR PHARMACOKINETICS

- When steady-state concentrations increase less than expected after a dosage increase, there are two typical explanations –

Saturable plasma protein

Auto induction.

PARMACOKINETIC MODELS

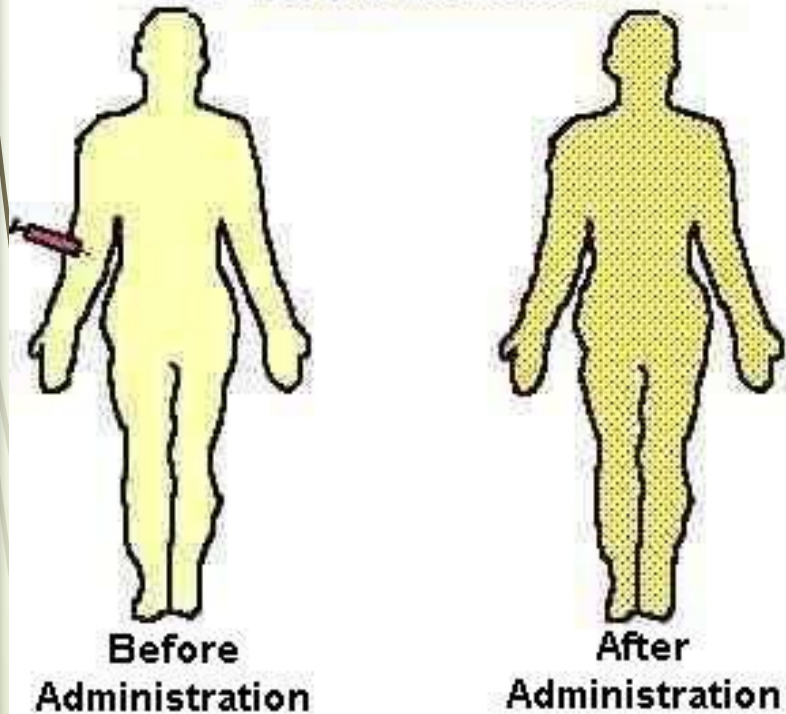
- A basic type of model used in pharmacokinetics is the compartmental model.
- Compartmental models are categorized by the number of compartments needed to describe the drug's behavior in the body.
- There are one- compartment, two-compartment, and multicompartment models.

PARMACOKINETIC MODELS

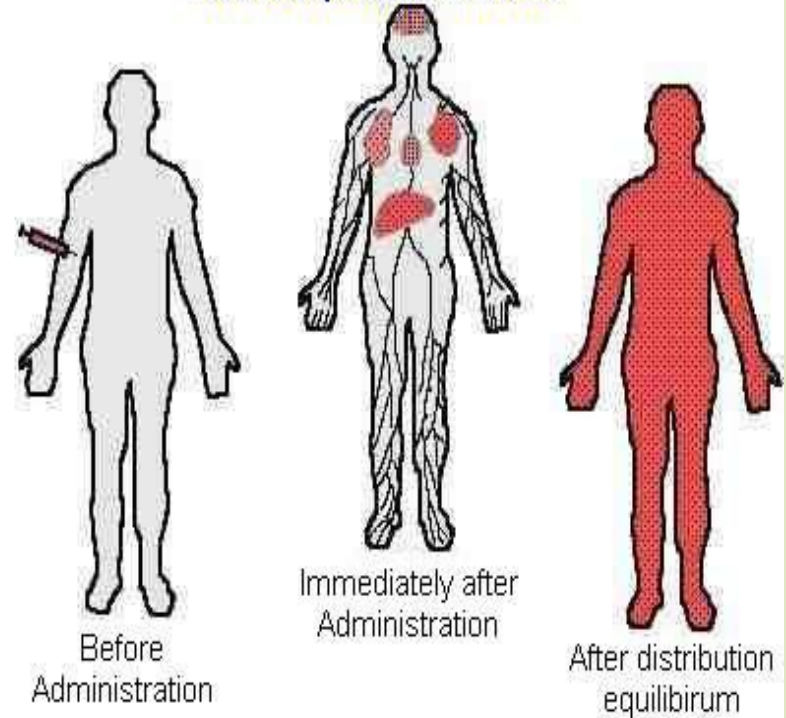
- The one compartment model assumes that the drug is evenly distributed throughout the body into a **single compartment**.
- This model is only appropriate for drugs which rapidly and readily distribute between the plasma and other body tissues.
- Drugs which exhibit a slow equilibration with peripheral tissues, are best described with a **two compartment model**.

PARMACOKINETIC MODELS

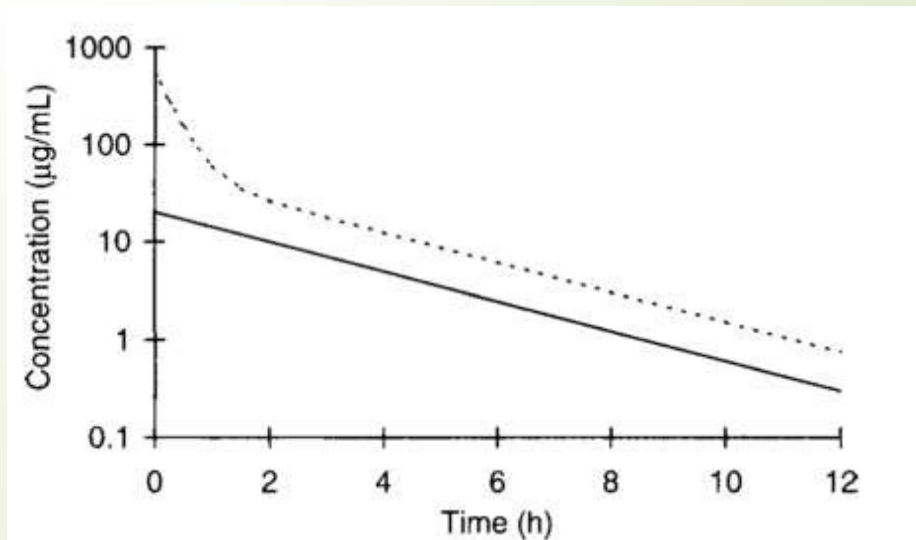
One compartment model



Two compartment model



PARMACOKINETIC MODELS



- The solid line shows the serum concentration/time graph for a drug that follows one-compartment model pharmacokinetics.
- The dashed line represents the serum concentration/time plot for a drug that follows two-compartment model pharmacokinetics after an intravenous bolus is given.

INDIVIDUALIZATION AND OPTIMIZATION OF DRUG THERAPY

Dosage regimen adjustment in Pediatrics

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- ADME is affected In the pediatric, due to a lack of full maturation of physiologic functions
 - individualized adjustment of medication and avoidance of certain medications altogether
- organ size and functional capacity
- Especial on drug disposition.
 - Substantial changes in organ function and enzyme processes

Dosage regimen adjustment in Pediatrics

- Frequent drug administration, for absence of weight adjustment
 - To compensate for more rapid body clearance observed in infants and children
 - Dose is adjusted by BSA

Dosage regimen adjustment in Pediatrics

- TBW is higher in pediatrics
 - 27-week-gestation premature infant may have 80% TBW vs the normal 60% for an adult
 - Vd increases
 - milligramper-kilogram basis is often much higher than that administered to older children and adults.

- ↓ Gastric acid secretion - pH ↑ - acidic drugs abs ↓, basic drugs abs ↑
- Immature biliary system – lipophilic drug abs ↓
- Vd ↑ acc. to their BW - water soluble drug distri ↑, fat soluble drug distri ↓
- Liver size ↑ - high dose preferred.
- Immature kidney - Toxicity ↑

► Dosage Calculation

Young's rule (For children 2 years and above):

$$\frac{(\text{Age (yr)})}{\text{Age (yr)} + 12} \times \text{adult dose}$$

$$\text{Age (yr)} + 12$$

Clark's rule:

$$\frac{(\text{Weight (lb)})}{150} \times \text{Adult dose}$$

$$150$$

Fried's rule (For infants upto 2 years old):

$$\frac{(\text{Age (month)})}{150} \times \text{adult dose}$$

$$150$$

Square meter surface area OR Mosteller's equation

$$\text{SA in m}^2 = \frac{(\text{height} \times \text{weight})^{1/2}}{60}$$

Dosage regimen adjustment in Geriatrics

- number of age-associated changes in ADME
 - increase in the body fat/water ratio (10–15% water volume decrement till age of 80.
 - fat-soluble drugs have an increased V_d
 - V_d of hydrophilic drugs therefore decreases

Dosage regimen adjustment in Geriatrics

- decrease in plasma protein
 - Increase the free drug concentration in plasma
 - Reduction of clearance
 - — Recommended to decrease the dose
- Reduction in GFR
 - Exacerbated by AKD or CKD
- Drug perfusion reduction^{AUC, S₁₀ P, n^{Clinical PK}}

Dosage regimen adjustment in Geriatrics

- drug-drug and drugfood interactions are common
- Most elder pts are confronted with ADR (2 to 3 times than adult)
- Absorption is not that much affected
- In frail elderly, drug metabolism is diminished
- Transdermal, subcutaneous and muscular tissue drug absorption reduction
 - Atrophy of the epidermis and dermis and reduced tissue blood perfusion
- injections should be avoided generally in this age group
 - erratic absorption and the high risk of sterile infiltrates.

PREGNANCY

- Weight gain – V_d $\uparrow$ - Loading dose should be high (important at rapid drug effect)
- Plasma albumin $\downarrow$ - unbound drug $\uparrow$ - acc – toxicity.
- GFR $\uparrow$ - clearance $\uparrow$ - higher maintenance dose
- Hepatic metabolism $\uparrow$ - due to enzyme induction – high maintenance dose
- Dose of a drug given at any stage should low as possible to minimize toxic effects to the foetus.
- Antidepressants & antipsychotics dose should be reduced slowly to parturition.

- Estimating renal function
 - An accurate estimation of renal function, or glomerular filtration rate (GFR), requires routine or repeated sophisticated techniques that are unsuitable for use. In practice, the serum creatinine concentration is used for day-to-day assessment of renal function. It has limitations, but it remains a robust and practical parameter for most clinical situations

Dosage regimen adjustment in renal impairment

- Patients with renal disease are treated with many medications, some of which may alter the absorption of other concomitantly administered drugs.
- Decrease plasma protein binding of many acidic drugs and increase in basic drugs
 - For acidic drugs, Hypoalbuminemia, conformational change of the binding sites on the albumin molecule and accumulation of endogenous substances which competitively displace acidic drugs from their binding sites on albumin
 - For basic drugs, elevated α_1 -acid glycoprotein

Dosage regimen adjustment in renal impairment

- The unbound fraction in plasma increase oral clearance of drugs and decrease in absorption

Dosage regimen adjustment in renal impairment

► Modifying of doses:

Decrease the drug dose and retain the usual dosage interval

Retain the usual dose and increase the dosage interval, or

Simultaneously decrease the dosage and prolong the dosage interval

DOSAGE REGIMEN ADJUSTMENT IN HEPATIC DISEASE

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- A decrease in liver first-pass effect results in extremely large increases in steady-state concentrations for orally administered drugs.

DOSAGE REGIMEN ADJUSTMENT IN HEPATIC DISEASE

- Liver plays a central role in ADME
 - liver blood flow, binding to plasma proteins, and biliary excretion depends on normal functioning of liver
 - Dosage adjustment in liver dysfunction
 - To avoid excessive accumulation of drug
- Liver is responsible for;
 - Elimination of exogenous substances including drugs by entero-hepatic circulation (biliary excretion), Metabolism (high no of metabolizing enzymes in liver and GI)
 - Poorly metabolized drugs high elimination half life

DOSAGE REGIMEN ADJUSTMENT IN HEPATIC DISEASE

- Liver dysfunction (Mainly Cirrhosis), other disease do not significantly affect PK
 - reduction in liver blood flow, the presence of intra- and extrahepatic portal-systemic shunting, a capillarization of the sinusoids and a reduction in the number and in the activity of the hepatocytes
 - clinical manifestations, such as esophageal varices, edema, ascites, severe impaired parenchymal function, and hepatic encephalopathy
- impaired production of albumin
- Impaired secretion of bile acids, bilirubin, and other organic anions
- intestine, the lungs, and the kidneys are also directly influenced
- contribute to alteration in CL_{int} , CL_{ph} , CL_{renal} , CL_{total}

DOSAGE REGIMEN ADJUSTMENT IN HEPATIC DISEASE

ABSORPTION

- Increase in BA, especially drugs with high hepatic extraction ratios
 - Reduced presystemic hepatic metabolism and clearance
 - Eg, Carvedilol should be administered one-fifth of the normal dosage
- Due to the shunt, there will not be first pass metabolism
 - Eg midazolam BA increases

Drug	Normal	Cirrhosis	Fold increase	Reference
Carvedilol	0.19	0.83	4.4	[18]
Chlormethiazole	0.10	1.16	11.6	[17]
Labetalol	0.33	0.63	1.9	[19]
Meperidine	0.48	0.87	1.8	[20]
Metoprolol	0.50	0.84	1.7	[21]
Midazolam	0.38	0.76	2.0	[22]
Morphine	0.47	1.01	2.1	[23]
Nifedipine	0.51	0.91	1.8	[24]
Nisoldipine	0.04	0.15	3.8	[25]
Pentazocine	0.18	0.68	3.8	[20]
Propranolol	0.36	0.60	1.7	[26]
Verapamil	0.10	0.16	1.6	[27]

DOSAGE REGIMEN ADJUSTMENT IN HEPATIC DISEASE

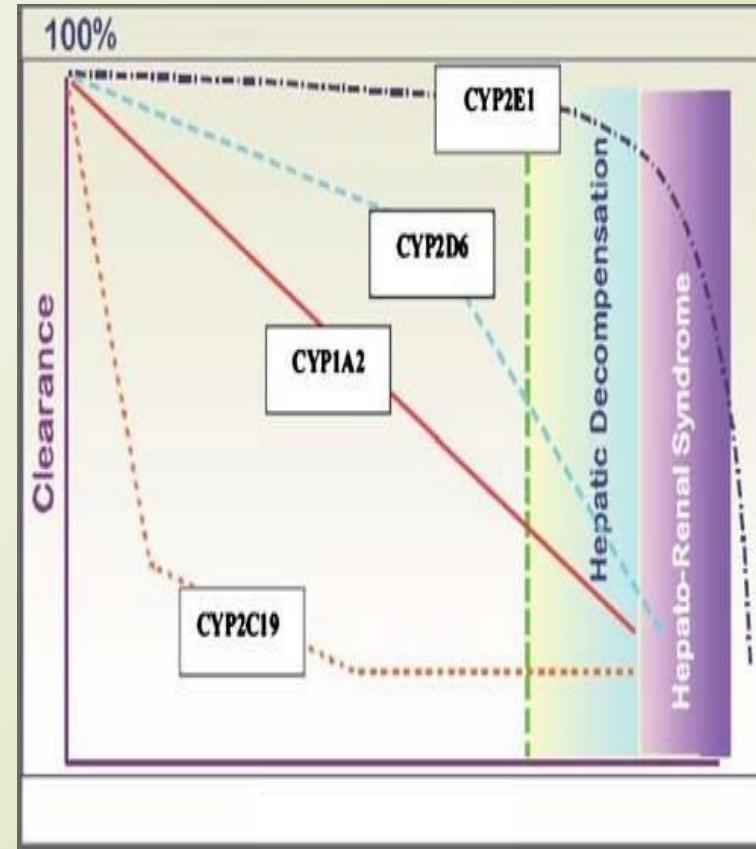
Plasma protein binding and distribution

- albumin or α 1-acid glycoprotein production decreases qualitatively and quantitatively
 - Vd increases for these protein bound drugs
- Accumulation of Billirubin Decreases drug-protein binding for certain drugs
- water-soluble drugs have higher Vd due to ascites

DOSAGE REGIMEN ADJUSTMENT IN HEPATIC DISEASE

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- **Elimination and Metabolism**
- Liver is the main organ in metabolism
 - Depends on metabolic enzyme activity and the activity of sinusoidal
- hepatobiliary drug disposition
- Liver dysfunction
 - reduction in absolute liver cell mass or a decrease in enzyme activity, sinusoidal capillarization, reduced liver perfusion, Reduced formation or secretion of bile into the duodenum
 - will lead to a decreased clearance of substances,



DOSAGE REGIMEN ADJUSTMENT IN HEPATIC DISEASE

Renal Excretion

- Advanced hepatic disease is commonly complicated by impaired renal function
 - mainly excreted in unchanged form by the kidneys such as the diuretics furosemide and bumetanide
 - Due to reduced muscle mass and impaired metabolism of creatine to creatinine in a number of patients with severe liver disease, estimations of creatinine clearance based on serum creatinine measurements is inaccurate
- increased fractional tubular secretion of creatinine in patients with cirrhosis as the glomerular filtration rate deteriorates

Dosage regimen adjustment in Obesity

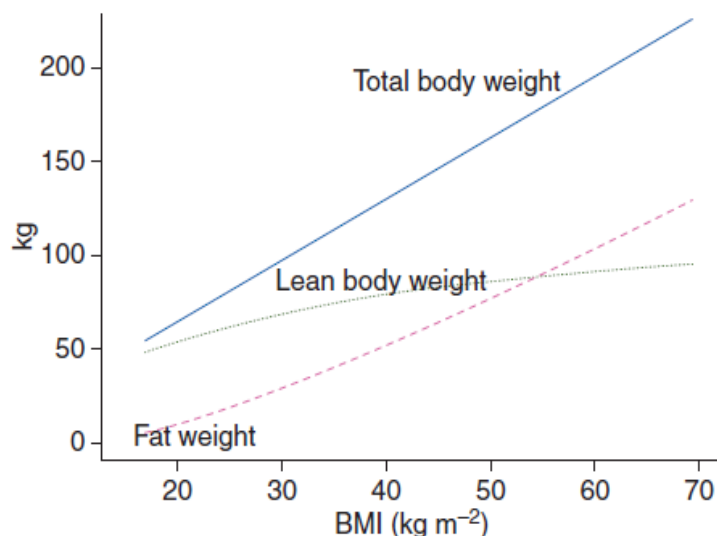
- The increasing worldwide incidence of obesity;
 - Burden on patient safety and financial perspective
- Numerous physiological changes occur
 - **Absorption**
 - **Ditribution**
 - **Metabolism**
 - **Excretion**
- Most smart profesionales reduce doses in obese patients
- Pharmacokinetics knowledge is important
 - **Narrow TW and anaesthetics**
Clinical PK

Dosage regimen adjustment in Obesity

Table 2. Terminologies used to describe different BMI categories

Classification (various terminologies)	BMI (kg/m ²)
Underweight	< 18.5
Normal weight	18.5 to <25
Overweight	≥ 25
Pre-obese	25 to <30
Obese	≥ 30
Obese (obesity class I)	30 to <35
Severely obese (obesity class II)	35 to <40
Morbidly obese (obesity class III)	≥ 40
Super obese	≥ 50
Super-super obese	≥ 60

- TBW, dosing recommendation for normal wt (TBW=LBW=IBW)
- Dosing in obese is best described by LBW.
 - TBW, will cause overdose and IBW cause underdose.



Dosage regimen adjustment in Obesity

Absorption

- Delayed gastric emptying
 - Resulting higher fat diet or gastric distension
 - result in a lower C_{max} or reduced absorption
- With an oral drugs where the absorption is increased by taking with a fatty meal
 - higher absorption, due to a presumptive higher intake of fatty foods
- IM drug administration, be administered deep subcutaneously

Dosage regimen adjustment in Obesity

Distribution

- It is measured by V_d
 - EBW is approximately 30% water in ECF
 - Increase V_d
 - barbiturates and benzodiazepines
- EBW is also mainly composed of fats in tissues
 - Increase V_d
- Also Depends on lipo and hydrophilicity of the drug
 - In obese pts, V_d of lipophilic drugs increase
- Other factors; Drug-protein binding, increased blood volume and cardiac output and poorer peripheral perfusion

Metabolism

- Hepatic volume increase, more likely due to fatty infiltration
 - cytochrome P450 (CYP2E1 and possibly CYP1A2 and CYP2C9) have raised levels, and CYP3A4 has lower levels

Dosage regimen adjustment in Obesity

Excretion

- The effect of obesity on renal function is bidirectional
 - baseline general increase in renal clearance
 - but the higher incidence of renal dysfunction, results in decreased renal function

General Principles

- General principles of prescribing in renal failure:
- *GOOD KNOWLEDGE OF PHARMACOKINETICS OF THE DRUGS USED IS NECESSARY!*
- Use drugs only if a definite indication is present
- Choose a drug with no/minimal nephrotoxicity
- Use recommended regimen given. If no data, use your COMMONSENSE!
 - •Use TDM to adjust doses
 - •Avoid prolonged courses of potentially toxic drugs
 - •Monitor patient for clinical effectiveness & toxicity

General Principles

Modification of dosage regimen:

- Look at the drug's renal & non renal elimination (%)
- How toxic the drug is
- Two approaches:
 - Standard dose -----Extended intervals
 - Reduced dose-----usual intervals.
- **LOADING DOSE REMAINS THE SAME!**

General Principles

- *For many drugs, an unchanged loading dose followed by half the usual adult maintenance dose is all that is done!*
- For more toxic drugs use dosage regimen based on GFR.
- Subsequent dosage based on:
 - Clinical response
 - Signs of toxicity
 - Plasma concentration

General Principles

Modification of dosage regimen:

- If antibiotic given in usual doses but at extended intervals-----there may be considerable periods of sub-inhibitory Concentrations.
- More frequent doses---toxicity!
- Measure plasma concentrations frequently.
 - Risk versus benefit

General Principles

- Following data is given in formularies, text books or leaflets:
 - Elimination $t_{1/2}$ ---both normal & in renal failure
 - Unchanged excretion (%)
 - Normal extra renal excretion
 - Normal dose interval (hours)
 - Dose adjustment for renal failure based on Creatinine clearance (ml/min): >50 , $10-50$, <10 .

General Principles

- **Limitations of the above data:**
 - Status of renal function should be determined not only before, but also the entire period of treatment
 - These dosage guidelines do not necessarily apply to elderly patients.
 - Many aspects of individual patients influence drug response in uremia
 - Monitor plasma concentration, especially for drugs with a low therapeutic index.
 - Monitor patients for unexpected drug toxicity.